

Jaime Manríquez

Computational Engineer | Numerical Simulation & Scientific Computing

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PROFILE

I am an applied mathematician with a PhD from Lund University (LTH) and experience in multiphysics simulation, from the underlying theory through to applied work. On the theoretical side I built a finite element library in MATLAB for coupled flow problems involving curved interfaces, including fluid-structure interaction and flow through porous media. On the applied side I wrote simulation software that predicts clogging in slow sand filters and their efficiency in removing pathogens from drinking water, working alongside microbiologists and water engineers. That project showed me how much I enjoy interdisciplinary work; the hard part was rarely the mathematics, it was getting all of us to explain ourselves plainly. I am looking for technical work that needs both.

TECHNICAL SKILLS

Numerical methods & modelling

- Discretization of partial differential equations: finite element methods; finite difference and finite volume schemes; unfitted / non-body-fitted discretisations; non-matching and hanging-node meshes.
- Continuum mechanics: fluid mechanics; convection–diffusion–reaction systems; multi-phase and free-boundary problems; fluid–structure interaction.
- Raw water treatment: biofilm growth, pathogen removal and microbial dynamics.

Programming & tooling

- MATLAB (expert), Python (proficient), Julia (proficient), C++ (working knowledge).
- **HPC**: Parallel computation on clusters; batch workflows and scaling of long-running simulations.
- **Development practices**: Git version control; Linux/Unix shell, build systems and remote workflows; $\LaTeX$ for technical documentation.

Languages

- **Spanish**: Native | **English**: Fluent (TOEFL 115/120) | **Swedish**: Conversational (A2 coursework)

SELECTED PROJECTS

Finite element library for interface problems on curved geometries

2020 – 2026

Main developer — MATLAB

- **Problem**: Coupled physical domains (fluid–structure, fluid–porous) normally require meshes conforming to the interface. For curved geometries this is expensive and often impractical, and mismatched subdomain meshes degrade accuracy.
- **Built**: A hybridizable discontinuous Galerkin (HDG) finite element library using discontinuous basis functions on unfitted meshes, combined with the Transfer Path Method to bridge geometry not captured by the mesh. Supports high-order elements, non-matching subdomain meshes and curved boundaries.
- **Result**: High-order (super)convergence preserved even with gaps between subdomain meshes, removing the need for conforming remeshing. Verified on 2D benchmarks, applied to Fluid–Fluid, Fluid–Medium and Fluid–Structure couplings.

Simulation software for biofilm growth and pathogen removal in water treatment

2021 – 2026

Main developer — MATLAB, Python, Julia

- **Problem**: Slow sand filtration is a widely used drinking-water treatment process whose performance depends on a biological layer operators cannot observe directly. Plant behaviour under changing temperature, load and flow was not predictable from existing models.
- **Built**: A multi-phase continuum simulator coupling biofilm growth, nutrient transport and pathogen removal. Implemented positivity-preserving finite-difference and discontinuous Galerkin solvers for the 1D full-depth and 2D upper-filter models, handling discontinuous flux and free boundaries.
- **Result**: Simulator reproduces pathogen removal efficiency under variable environmental conditions, giving a predictive tool for filter performance. Developed with water-sector collaborators; published across three papers.

EDUCATION

PhD in Applied Mathematics

2021 – May 2026

LTH, Lund University

Lund, Sweden

- **Thesis:** *Modelling slow sand filtration and solving interface problems with the Transfer Path Method.*

MSc / Professional Degree, Engineering Mathematics

2015 – 2020

Universidad de Concepción

Concepción, Chile

- *Premio Universidad de Concepción 2020*, top graduating student in Engineering Mathematics.

PROFESSIONAL EXPERIENCE

Doctoral Researcher

2021 – May 2026

Centre for Mathematical Sciences, Lund University

Lund, Sweden

- Collaborated with domain specialists outside mathematics (water engineering, microbiology)
- Secured competitive project funding through written proposals; presented at seven international conferences, including as invited speaker.
- Taught exercise seminars in numerical methods for differential equations, numerical analysis for electrical engineers, optimisation and analysis in one and several variables; supervised student modelling projects and developed course material.

Modelling Intern

2020

ATCAS S.A.

Chile

- Developed a predictive model for forest fire spread, translating terrain, fuel-load and meteorological inputs into a computational description of front propagation.

Engineering Intern

2019

CADETECH

Chile

- Built a non-linear optimisation model for mooring-chain placement on water vessels, formulating the physical layout constraints as a constrained optimisation problem and solving it numerically.

Teaching Assistant

2016 – 2020

Universidad de Concepción

Concepción, Chile

- Taught exercise seminars in numerical analysis, fluid mechanics, calculus I–III and real analysis to engineering cohorts.
- Constructed course material and examinations, including for fluid mechanics for metallurgical engineers.
- Produced and edited video material for online teaching support in the context of the IDECLab pedagogical project.

PUBLICATIONS & PROFESSIONAL ACTIVITIES

- **Four peer-reviewed journal publications** (one further under review) in computational mechanics and numerical analysis: *CMAME* (2022), *IMA J. Numerical Analysis* (2025), *Applied Mathematical Modelling* (2025), *Computers & Mathematics with Applications* (2026).
- **Leadership & service:** Organiser of the PhD seminar series; elected member of PhD student council, Lund University (2025); departmental board representative (2022).